

I Theoretical Questions

1) (a) Formula: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$

Month	Calculation	Return
1	$(108 - 100) / 100$	$0.08 = 8\%$
2	$(103 - 108) / 108$	$-0.0463 = -4.63\%$
3	$(115 - 103) / 103$	$0.1165 = 11.65\%$
4	$(110 - 115) / 115$	$-0.0435 = -4.35\%$

(b) Log Returns:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

Month	Log Return
1	$\log (108/100) = 0.07696 = 7.70\%$
2	$\log (103/108) = -0.0474 = -4.74\%$
3	$\log (115/103) = 0.1102 = 11.02\%$
4	$\log (110/115) = -0.0445 = -4.45\%$

(c) month	$ R_t - r_t $
1	0.0030
2	0.0010
3	0.0065 $\Rightarrow$ least accurate in
4	0.0009

Month (3)
(Largest Return)

$$\underbrace{\ln(1+x)}_{r_t} \approx \underbrace{x}_{R_t} - \frac{x^2}{2} \quad \text{(Taylor Series Expansion)}$$

(if R_t is very small)

2) (a) linearity of expectation

$$\Rightarrow E[\underbrace{w_1 R_1 + w_2 R_2}_{R_p}] = w_1 E[R_1] + w_2 E[R_2]$$

$$E[R_p] = 0.4 \times 15\% + 0.6 \times 7\% \\ = 6\% + 4.2\% = 10.2\%$$

(b) No, it does not change. Linearity of expectation always holds, regardless of how R_1 & R_2 are correlated.

$\Rightarrow$ Correlation affects the variance of the portfolio, not its expected return.

$$3) \sigma_p^2 = \text{Var}(w_1 R_1 + w_2 R_2)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\sigma_p^2 = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2\text{Cov}(w_1 R_1, w_2 R_2)$$

$\rightarrow$ Use $\text{Var}(cX) = c^2 \text{Var}(X)$
 $\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

$\rightarrow \text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$

$$\boxed{\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2}$$

$$\sigma_p^2 = (0.5)^2 (25)^2 + (0.5)^2 (12)^2 + 2(0.5)(0.5) \rho (25)(12)$$

$$= 156.25 + 36 + 150\rho$$

4) (a) $\rho = +1 \Rightarrow \sigma_p^2 = 342.25$
 $\sigma_p = 18.5\%$

$\rho = 0 \Rightarrow \sigma_p^2 = 192.25$
 $\sigma_p = 13.87\%$

$\rho = -1 \Rightarrow \sigma_p^2 = 42.25$
 $\sigma_p = 6.5\%$

(b) σ_p = weighted avg. when $\rho = +1$

Proof: - when $\rho = 1$

$$\sigma_p = \sqrt{w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B}$$

$$= \sqrt{(w_A \sigma_A + w_B \sigma_B)^2} = w_A \sigma_A + w_B \sigma_B$$

Numerically: $0.5 (25) + 0.5 (12) = 18.5\%$

(c) $\rho = -1, \sigma_p = 0$

$$w_A^* \sigma_A = (1 - w_A^*) \sigma_B \Rightarrow w_A^* (25) = (1 - w_A^*) 12$$

$$\Rightarrow 37 w_A^* = 12$$

$$w_A^* = \frac{12}{37} \approx 0.3243$$

5) Portfolio Variance $\Rightarrow \sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

$\rho_{12} < 1 \Rightarrow$ cross term is smaller compared to $\boxed{\rho_{12} = 1}$ case

correlation $\Rightarrow \sigma_p^2 < (w_1 \sigma_1 + w_2 \sigma_2)^2$

$\Rightarrow$ portfolio risk is less than the weighted average of individual risks.

$\Rightarrow$ Intuitively, the assets do not move perfectly together.

$\Rightarrow$ One asset may rise while another falls or stays stable.

Thus, diversification lowers risk because losses in one asset can be partially offset by gains in another.

6) (a) $\Sigma = \begin{bmatrix} \rho_{11} \sigma_1 \sigma_1 & \rho_{12} \sigma_1 \sigma_2 & \rho_{13} \sigma_1 \sigma_3 \\ \rho_{21} \sigma_2 \sigma_1 & \rho_{22} \sigma_2 \sigma_2 & \rho_{23} \sigma_2 \sigma_3 \\ \rho_{31} \sigma_3 \sigma_1 & \rho_{32} \sigma_3 \sigma_2 & \rho_{33} \sigma_3 \sigma_3 \end{bmatrix}$

$= \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}$

(b) Symmetric $\Rightarrow$ as $\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j = \rho_{ji} \sigma_j \sigma_i = \Sigma_{ji}$

$$7) \text{ Sharpe Ratio} = \frac{E[R_p] - R_f}{\sigma_p}$$

$$R_f = 4\%$$

$$S_x = \frac{14 - 4}{22} \approx 0.455$$

$$S_y = \frac{9 - 4}{10} = 0.5$$

$$S_z = \frac{20 - 4}{35} \approx 0.457$$

$$(b) \text{ Ranking} \Rightarrow Y > Z > X$$

(Y) $\Rightarrow$ Highest Sharpe Ratio

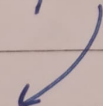
$\Rightarrow$ Highest return per unit Risk

(c) Absolute return $\Rightarrow E[R_p]$

Prefer Y over Z

Because lower risk and hence higher Sharpe Ratio.

8) (a) GMV = portfolio which achieves the lowest possible variance across the entire feasible set.



 Risk

$\Rightarrow$ Left most tip of Min. Var. frontier ($\sigma_p, E[R_p]$) space because no other feasible portfolio can have smaller risk.

(b) Incorrect.

→ Every portfolio on the lower half of the min. Var. frontier is dominated: $\exists$ another portfolio with same σ_p but a higher $E[R_p]$ on the upper half.

No rational investor accepts a lower return for the same level of risk. A very risk-averse investor should hold the GMV portfolio, or something on the efficient (upper) frontier → never below it.

(c) CML $\Rightarrow$ Straight line drawn from risk free rate $(0, R_f)$ tangent to the efficient frontier.

$$\text{Slope of CML} = \frac{E[R_m] - R_f}{\sigma_m}$$

$\Downarrow$

Sharpe Ratio of market portfolio.

g) (a) $SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}} = \frac{0.3}{\sqrt{36}} = 5\%$

5% $\Rightarrow$ large uncertainty

expected returns for most assets lie roughly in 8-15% range so $\pm 5\%$.

Error makes historical estimates quite unreliable. MVO $\Rightarrow$ Highly sensitive to noisy inputs.

(b) MVO $\Rightarrow$ Error maximiser $\Rightarrow$

because it tends to assign large weights to assets whose $\hat{\mu}$ happens to be overestimated due to sampling noise. This, amplifies the estimation errors.

Using ML - predicted returns (from regression models, gradient Boosting)

provides smoother, forward looking estimates that are less prone to overfitting noisy historical data,

(c) $\Sigma \Rightarrow$ symmetric

$$\text{Unique parameters} = \underbrace{N}_{\text{Variances}} + \underbrace{\frac{N(N-1)}{2}}_{\text{Covariances}}$$

$$= \frac{N(N+1)}{2}$$

For $N=50$,

$$\frac{50 \times 51}{2} = \boxed{1275}$$

⇒ 50 variances
1225 unique covariances



Large no. of parameters

⇒ Thus, estimating them requires huge amount of data.

With limited history, sample covariance matrix becomes unreliable.



causing serious numerical problems in optimisation.